



As novidades do Flutter?

Tudo o que rolou no Google IO



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GDE Flutter & Dart
Microsoft MVP
@toshiossada

Toshi Ossada

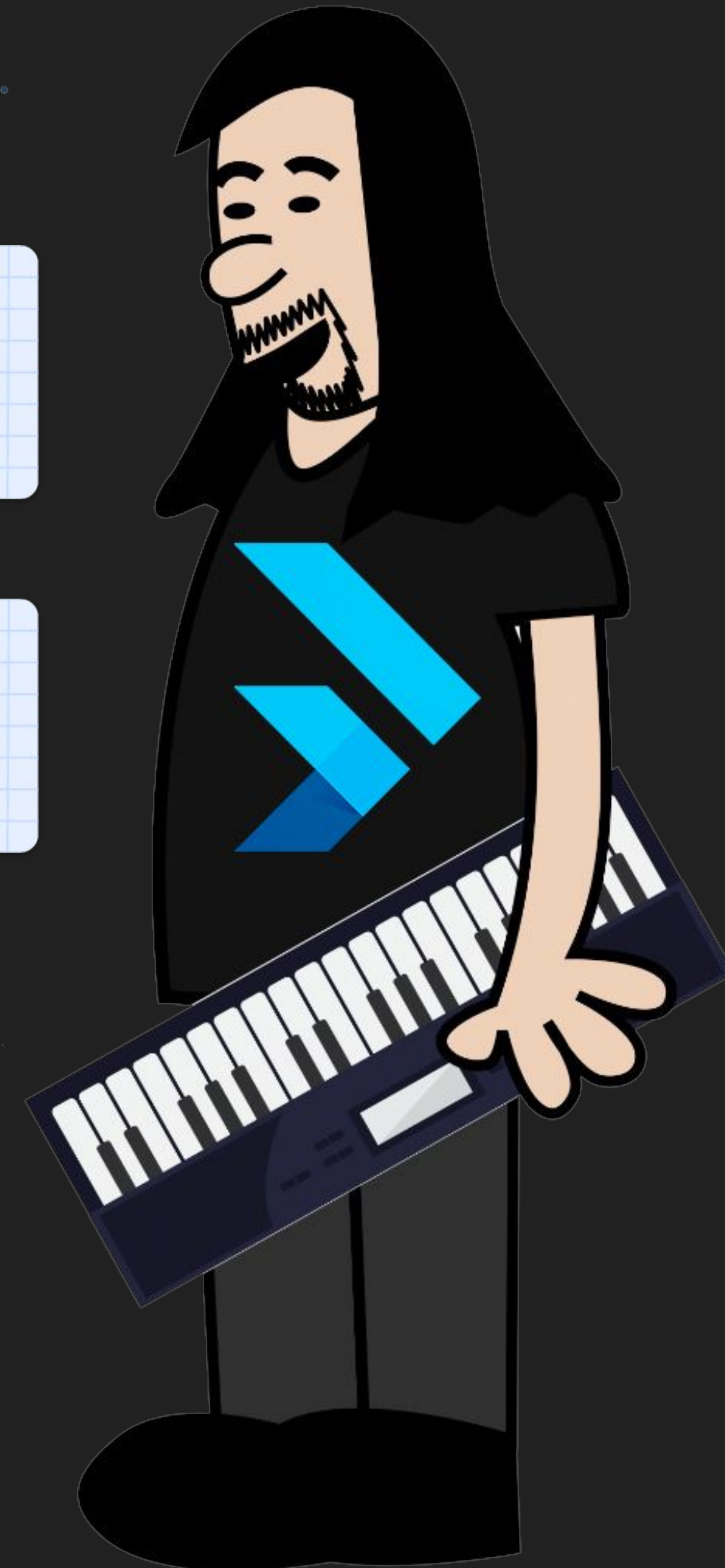
toshiossada.dev



flutterbrasil.com.br



GDG





```
function filterStudies({ studies, filterByOrg = false, filterByLeadOrganizat
```

~~“Aonde você quer rodar seu app?”~~

**“O que você quer
desenvolver?”**

#1 most used

multiplatform app framework

28% of new apps

“Apptopia tracks millions of apps in the Apple AppStore and Google Play Store, and analyzes and detects which developer SDKs were used to create the apps. Flutter is one of the most popular SDKs we track: In the Apple App Store it has grown steadily in usage from around 10% of all tracked free apps in 2021 to nearly 30% of all tracked free apps in 2024!”

— Apptopia Inc.

Community

```
function filterStudies({ studies, filterByOrg = false, filterByLeadOrganizat
```


Flutter Meetup Network

- 150 groups
- 66 countries
- 83,124 members
- 541 events





843

contributors to Flutter GitHub repos

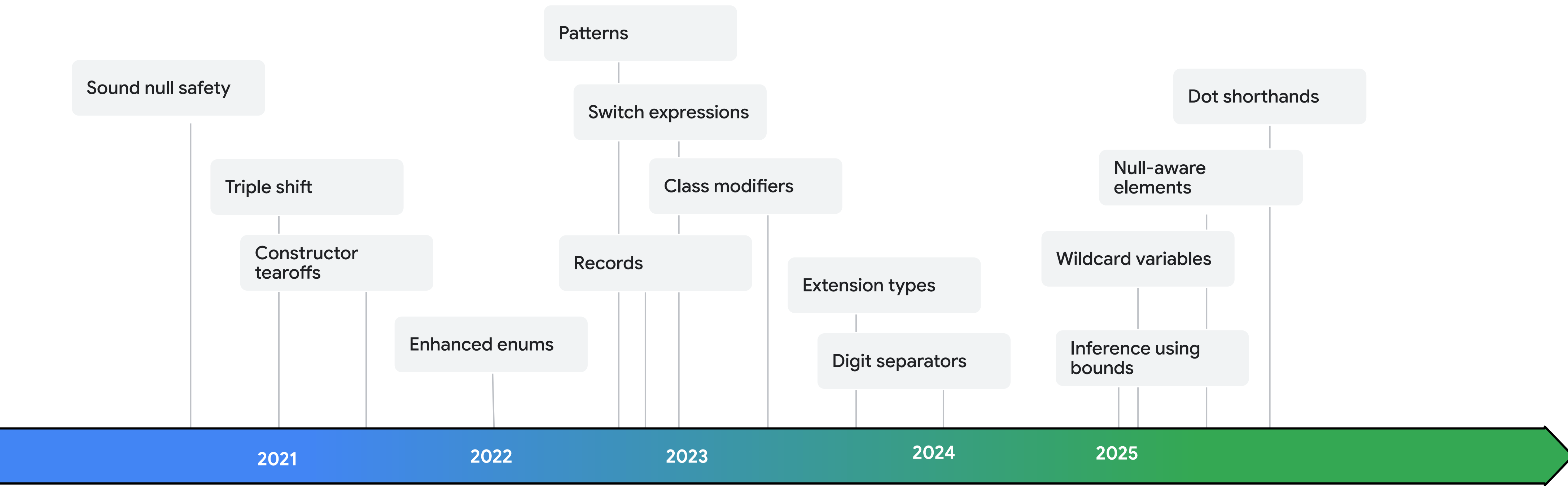
```
function filterStudies({ studies, filterByOrg = false, filterByOrgs = [], filterByOrgsRegex = [] }) {  
  return studies.filter(study => {  
    if (filterByOrg) {  
      return filterByOrgs.includes(study.organization) || filterByOrgsRegex.some(regex => regex.test(study.organization));  
    }  
    return true;  
  });  
}
```



flutter.dev/go/roadmap

#1 – Dart language

Evolução continua



Digit separators

Dart 3.6

Before

```
var v1 = 1000000;           // One million
var v2 = 0x0001422012345;    // MAC address
var v3 = 5551234567;         // US phone number
```

After

```
var v1 = 1_000_000;          // One million
var v2 = 0x00_14_22_01_23_45; // MAC address
var v3 = 555_123_4567;       // US phone number
```

Wildcard variables

Dart 3.7

Before

```
ListView.builder(  
  itemBuilder: (_, __) => ListTile(title: Text('foo'))  
)
```

After

```
ListView.builder(  
  itemBuilder: (_, _) => ListTile(title: Text('foo'))  
);
```

Dot shorthands

Future release

Before

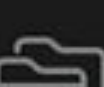
```
Column(  
  crossAxisAlignment: CrossAxisAlignment.stretch,  
  mainAxisAlignment: MainAxisAlignment.min,  
  children: [...],  
);
```

After

```
Column(  
  crossAxisAlignment: .stretch,  
  mainAxisAlignment: .min,  
  children: [...],  
);
```


#2 - Developer Experience

Editor de Propriedades



FLUTTER PROPERTY EDITOR

Icon
Creates an icon.

 Filter  

 double? **size**
null

 double? **fill**
null

 double? **weight**
null

 double? **grade**
null

 double? **opticalSize**
null

 String? **semanticLabel**
null

main.dart

lib > main.dart > _MyHomePageState > build

mainaziesize

No results

```
56 class _MyHomePageState extends StatelessWidget {
71   Widget build(BuildContext context) {
72     // by the _incrementCounter method above.
74     //
75     // The Flutter framework has been optimized to make rerunning build methods
76     // fast, so that you can just rebuild anything that needs updating rather
77     // than having to individually change instances of widgets.
78     return Scaffold(
79       appBar: AppBar(
80         // TRY THIS: Try changing the color here to a specific color (to
81         // Colors.amber, perhaps?) and trigger a hot reload to see the AppBar
82         // change color while the other colors stay the same.
83         backgroundColor: Theme.of(context).colorScheme.inversePrimary,
84         // Here we take the value from the MyHomePage object that was created by
85         // the App.build method, and use it to set our appBar title.
86         title: Text(''),
87       ),
88       body: Center(
89         child: Column(
90           mainAxisAlignment: MainAxisAlignment.start,
91           mainAxisAlignment: MainAxisAlignment.min,
92           crossAxisAlignment: CrossAxisAlignment.center,
93           children: <Widget>[
94             const Text('You have pushed the button this many times:'),
95             Text(
96               '$_counter',
97               style: Theme.of(context).textTheme.headlineMedium,
98             ),
99           ],
100         ),
101       ),
102     );
103   }
104 }
```


Flutter DevToolsFlutter InspectorPerformanceCPU ProfilerMemoryDebuggerNetwork

Select Widget ModeShow Implementation Widgets

Widget Tree

Switch

SizedBox

Divider

SizedBox

SettingsAbout

SettingsLink

InkWell

Padding

Row

Icon

Flexible

Padding

Text: "About Flutter Gallery"

SettingsFeedback

SettingsLink

InkWell

Padding

Row

Icon

Flexible

Padding

Text: "Send feedback"

SizedBox

Divider

Widget propertiesFlex explorer

Icon

h=24.0w=24.0

727.0

20.0

Row

height: 24.0

width: 24.0

bottom padding: 20.0

right padding: 727.0

icon

IconData(U+0E260)

size

24.0

color

#80ffffff

dependencies

[Directionality, IconTheme]

fill

defaultnull

weight

defaultnull

grade

defaultnull

opticalSize

defaultnull

shadows

defaultnull

semanticLabel

defaultnull

Console

Icon

IconData(U+0E260)

size: 24.0

fill: nulldefault

weight: nulldefault

grade: nulldefault

opticalSize: nulldefault

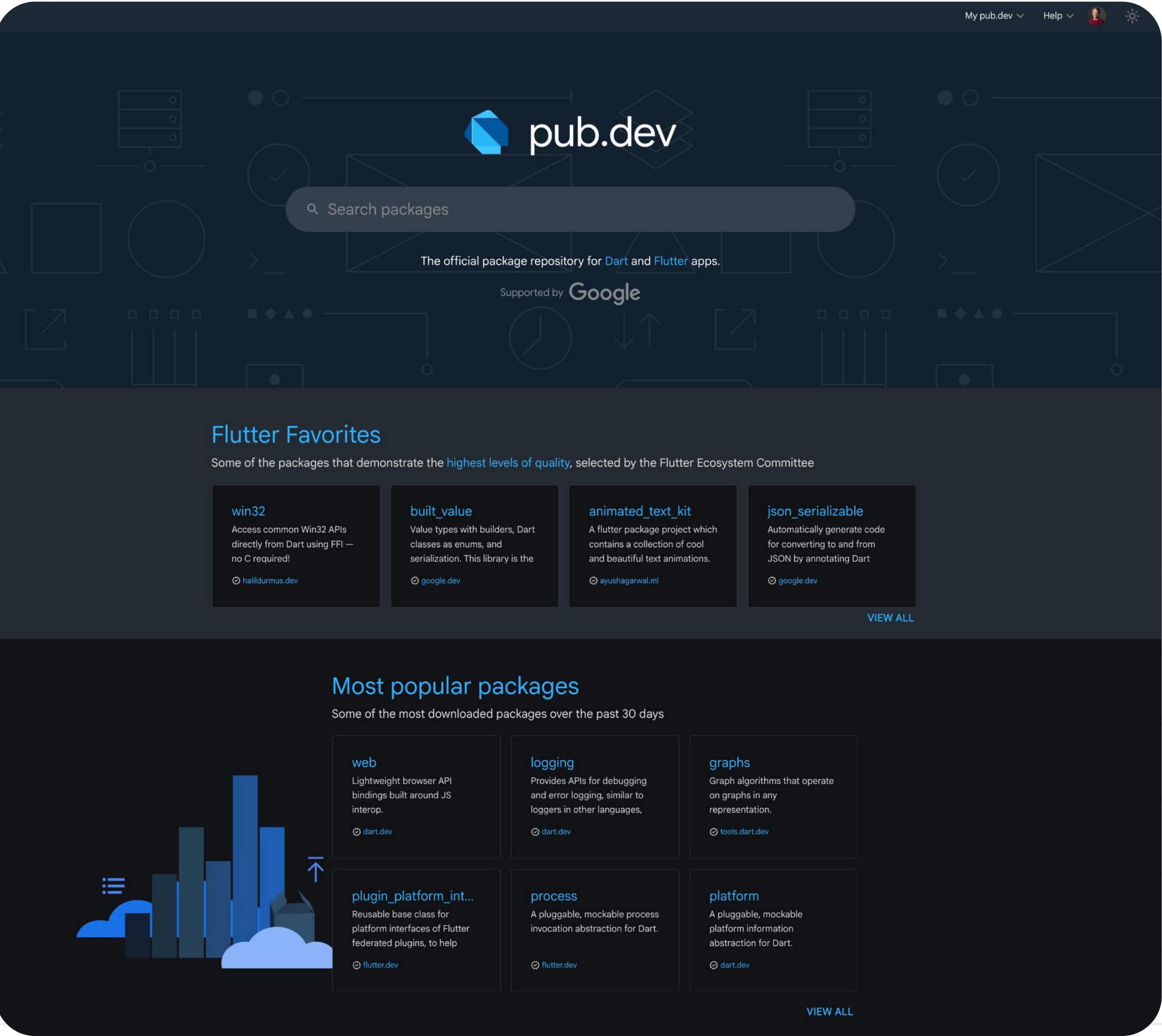
Read docs

Watch tutorial

x64 (64 bit) macos

pub.dev

Dark mode

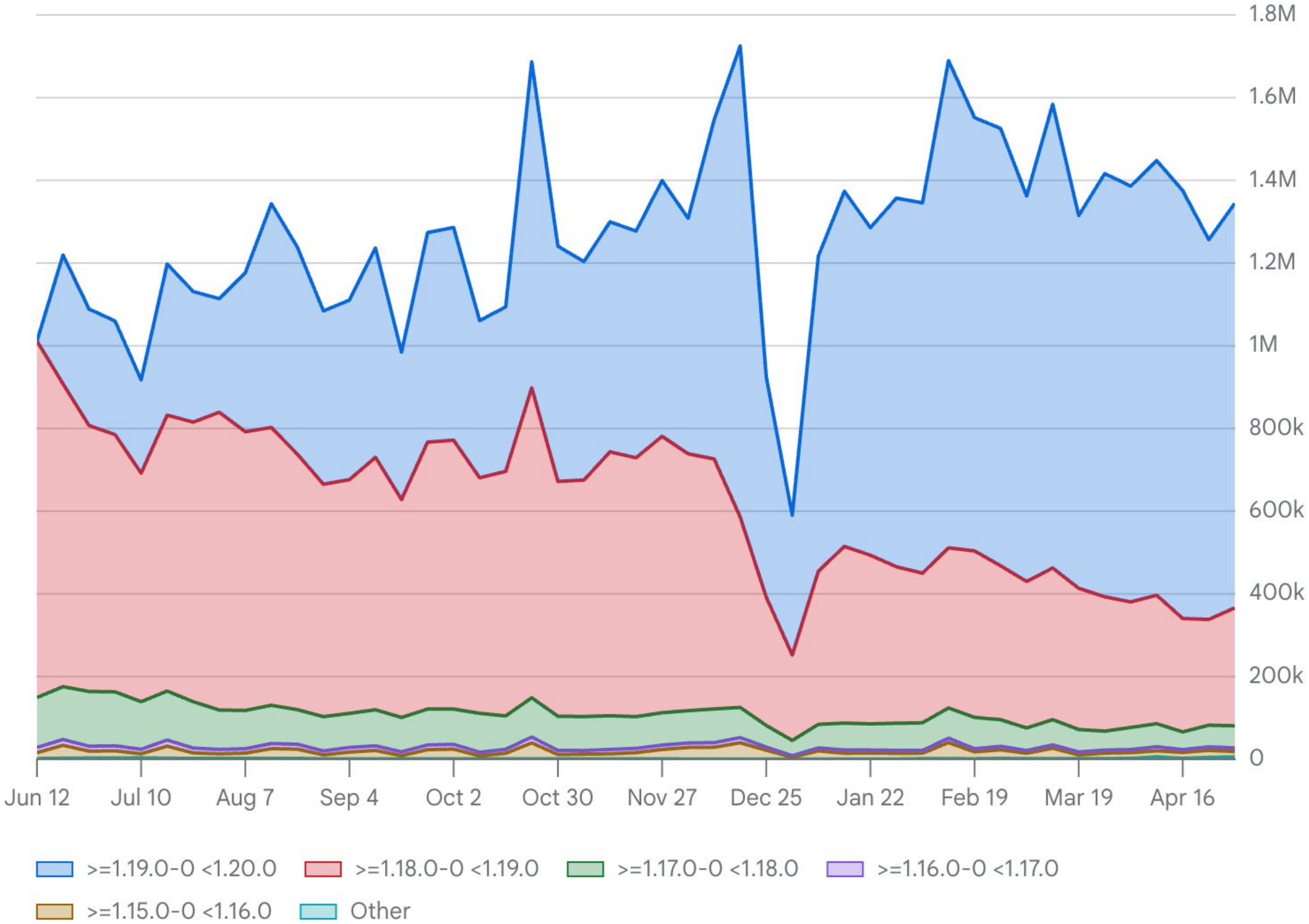


Contadores de download

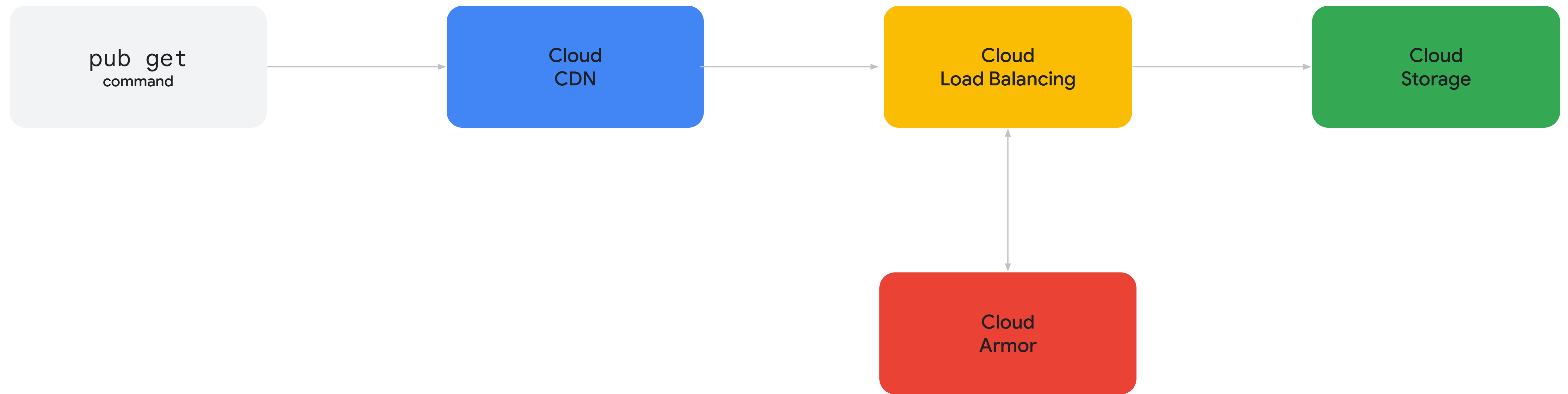
Weekly downloads

Display as: ☐ Unstacked ☒ Stacked ☐ Percentage

By versions: ☐ Major ☒ Minor ☐ Patch



pub.dev: mais poder na infraestrutura Cloud



Formatador Dart

Before

```
function(  
  name:  
    (param) => another(  
      argument1,  
      argument2,  
    ),  
);  
TabBar(tabs: [Tab(text: 'A'), Tab(text: 'B')],  
labelColor: Colors.white70);
```

After

```
function(  
  name: (param) => another(  
    argument1,  
    argument2,  
  ),  
);  
  
TabBar(  
  tabs: [  
    Tab(text: 'A'),  
    Tab(text: 'B'),  
  ],  
  labelColor: Colors.white70,  
);
```


Formatador Dart

analysis_options.yaml

```
formatter:
```

```
  trailing_commas: preserve
```

Cross-compilation



Much faster dart CLI tool

```
$ dart format
Formatted 93 files in 1.18 seconds

$ dart doc
Documented 1 library in 106.9 seconds

$ dart analyze
Analyzed 1 library in 19.0 seconds
```

Before

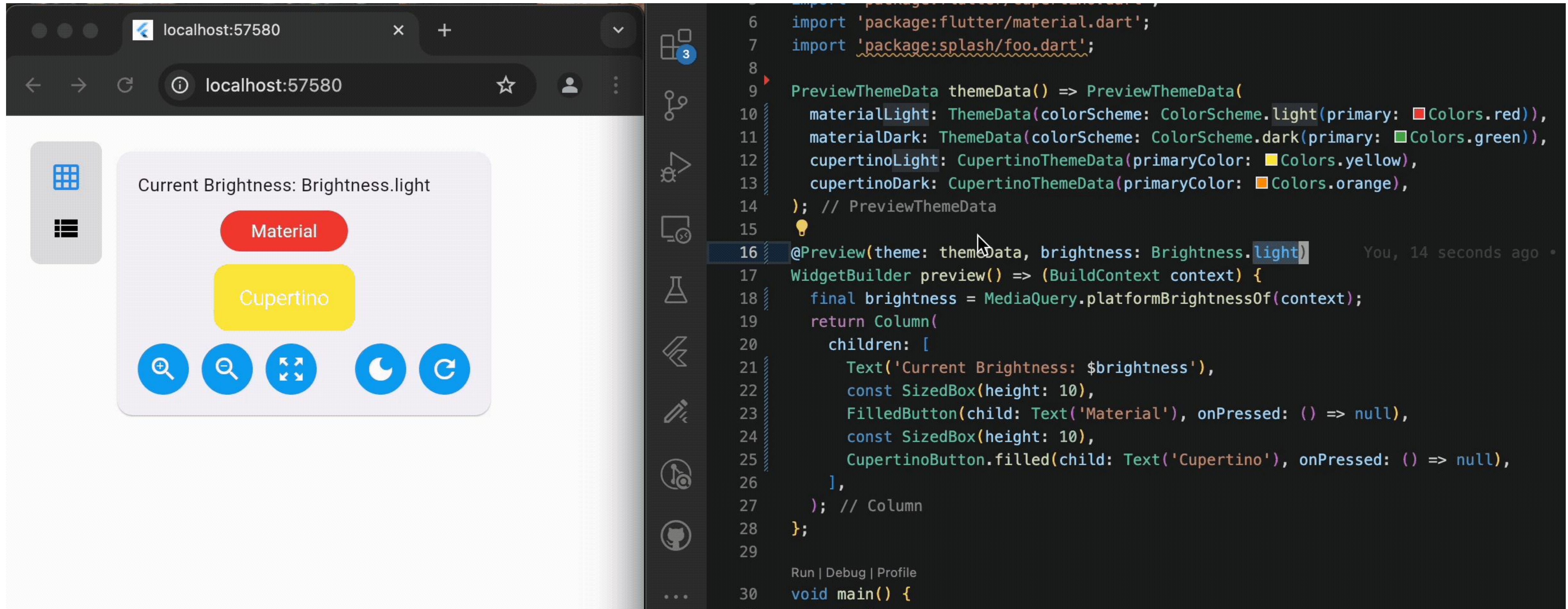
```
$ dart format
Formatted 93 files in 0.21 seconds

$ dart doc
Documented 1 library in 67.4 seconds

$ dart analyze
Analyzed 1 library in 12.8 seconds
```

After
(Future release)

Widget Previews



The image displays a development environment with a widget preview on the left and Dart code on the right.

Widget Preview (Left):

- Browser address bar: localhost:57580
- Current Brightness: Brightness.light
- Material theme button (red)
- Cupertino theme button (yellow)
- Interaction icons: zoom in, zoom out, pan, dark mode, refresh

Dart Code (Right):

```
import 'package:flutter/cupertino.dart';
import 'package:flutter/material.dart';
import 'package:splash/foo.dart';

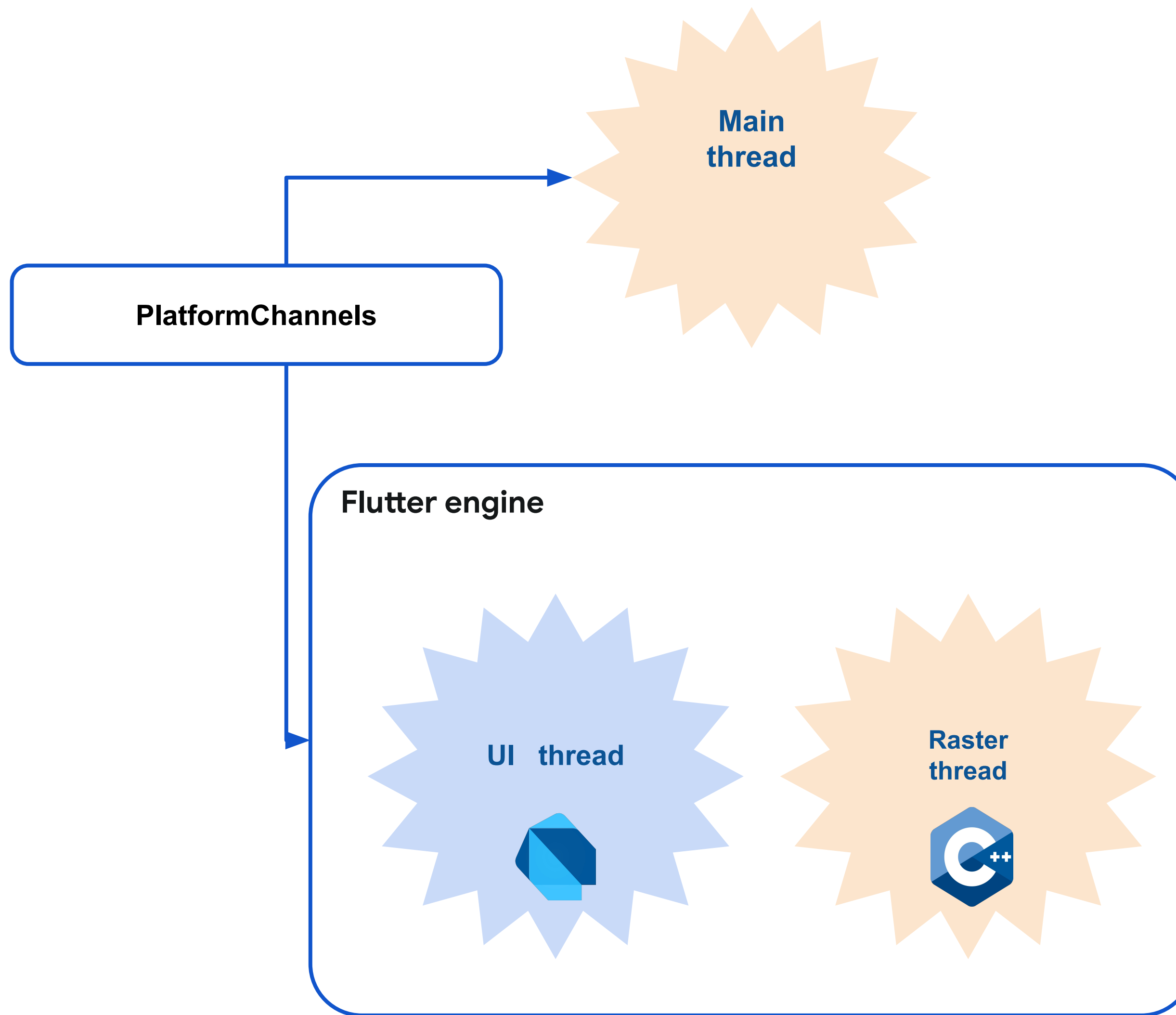
PreviewThemeData themeData() => PreviewThemeData(
  materialLight: ThemeData(colorScheme: ColorScheme.light(primary: Colors.red)),
  materialDark: ThemeData(colorScheme: ColorScheme.dark(primary: Colors.green)),
  cupertinoLight: CupertinoThemeData(primaryColor: Colors.yellow),
  cupertinoDark: CupertinoThemeData(primaryColor: Colors.orange),
); // PreviewThemeData

@Preview(theme: themeData, brightness: Brightness.light)
WidgetBuilder preview() => (BuildContext context) {
  final brightness = MediaQuery.platformBrightnessOf(context);
  return Column(
    children: [
      Text('Current Brightness: $brightness'),
      const SizedBox(height: 10),
      FilledButton(child: Text('Material'), onPressed: () => null),
      const SizedBox(height: 10),
      CupertinoButton.filled(child: Text('Cupertino'), onPressed: () => null),
    ],
  ); // Column
};

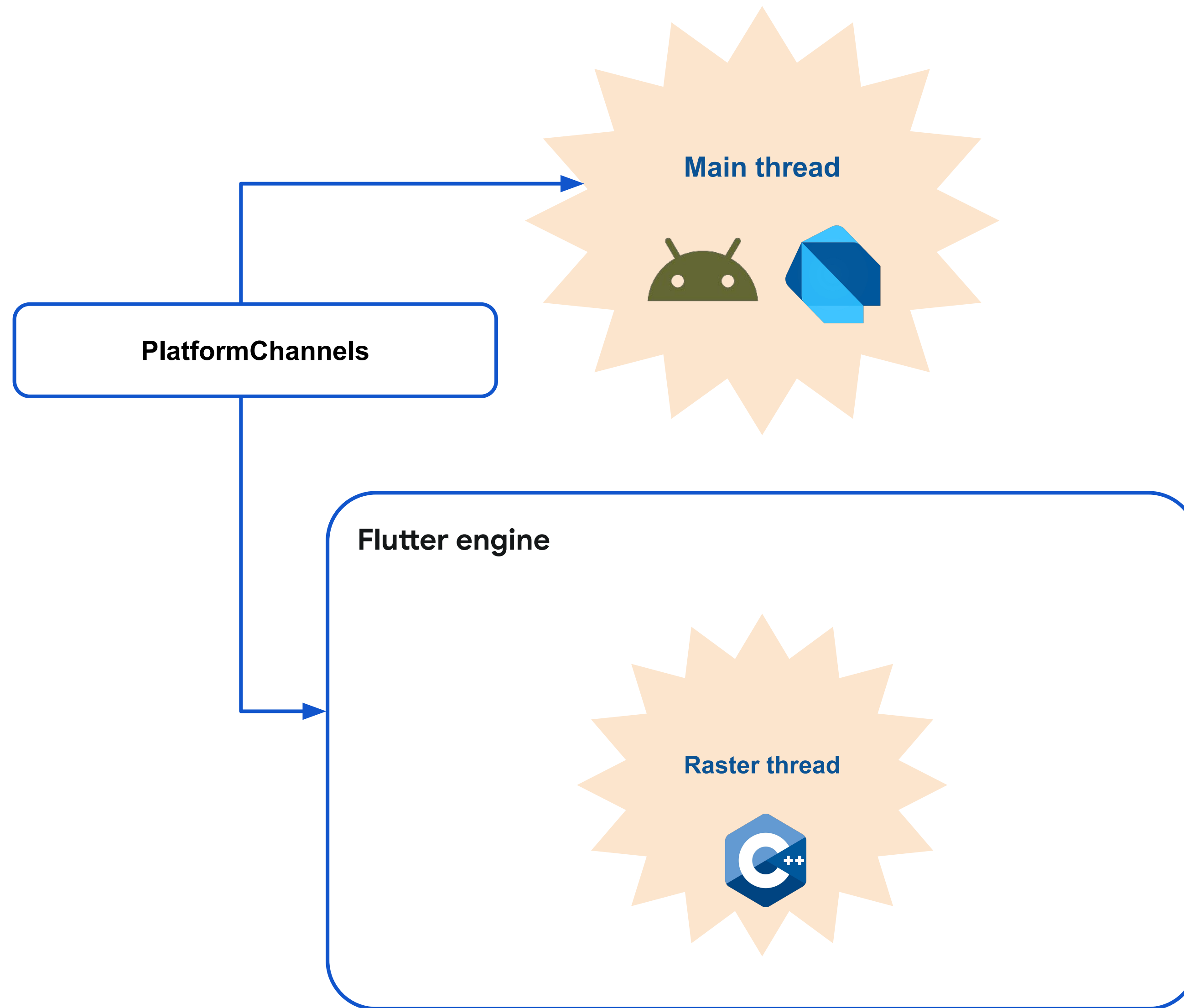
void main() {
```


#3 – Acesso as API Nativas

Native app



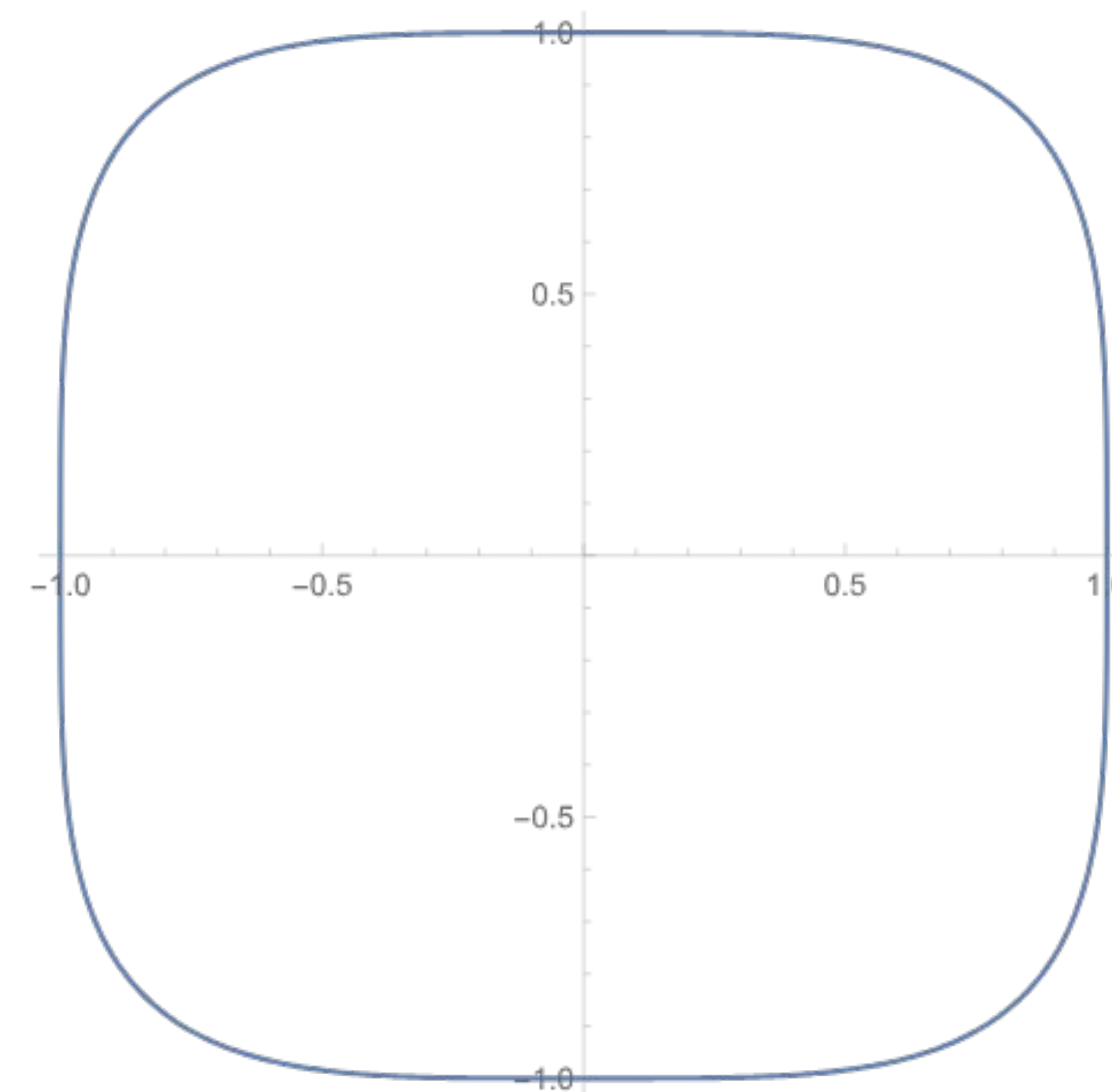
Native app



#3 – Melhorias para iOS

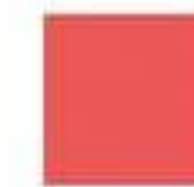
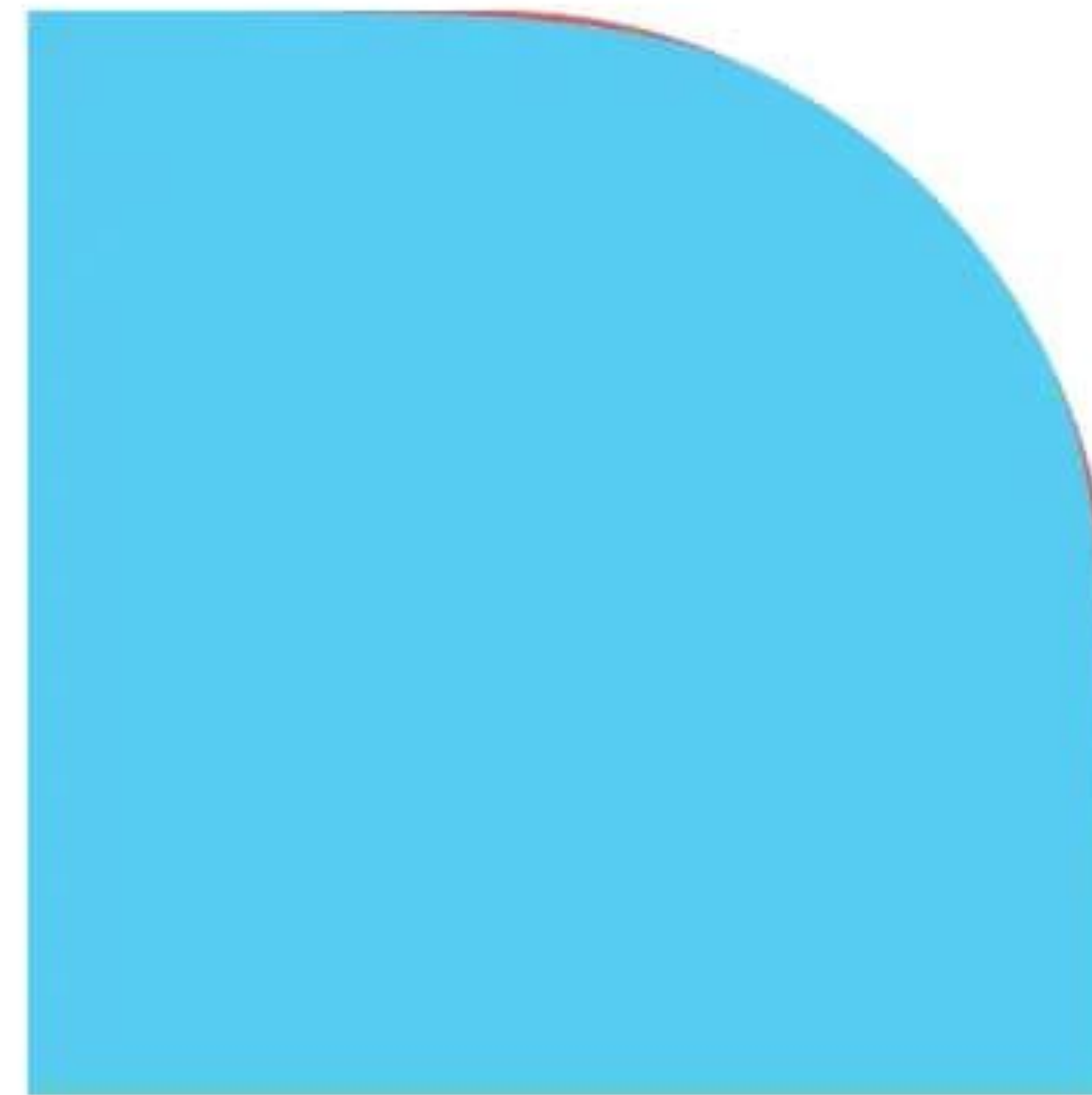


Squircles



Squircle vs Rounded Square

- Curvatura mais longa
- Transições de curva suaves



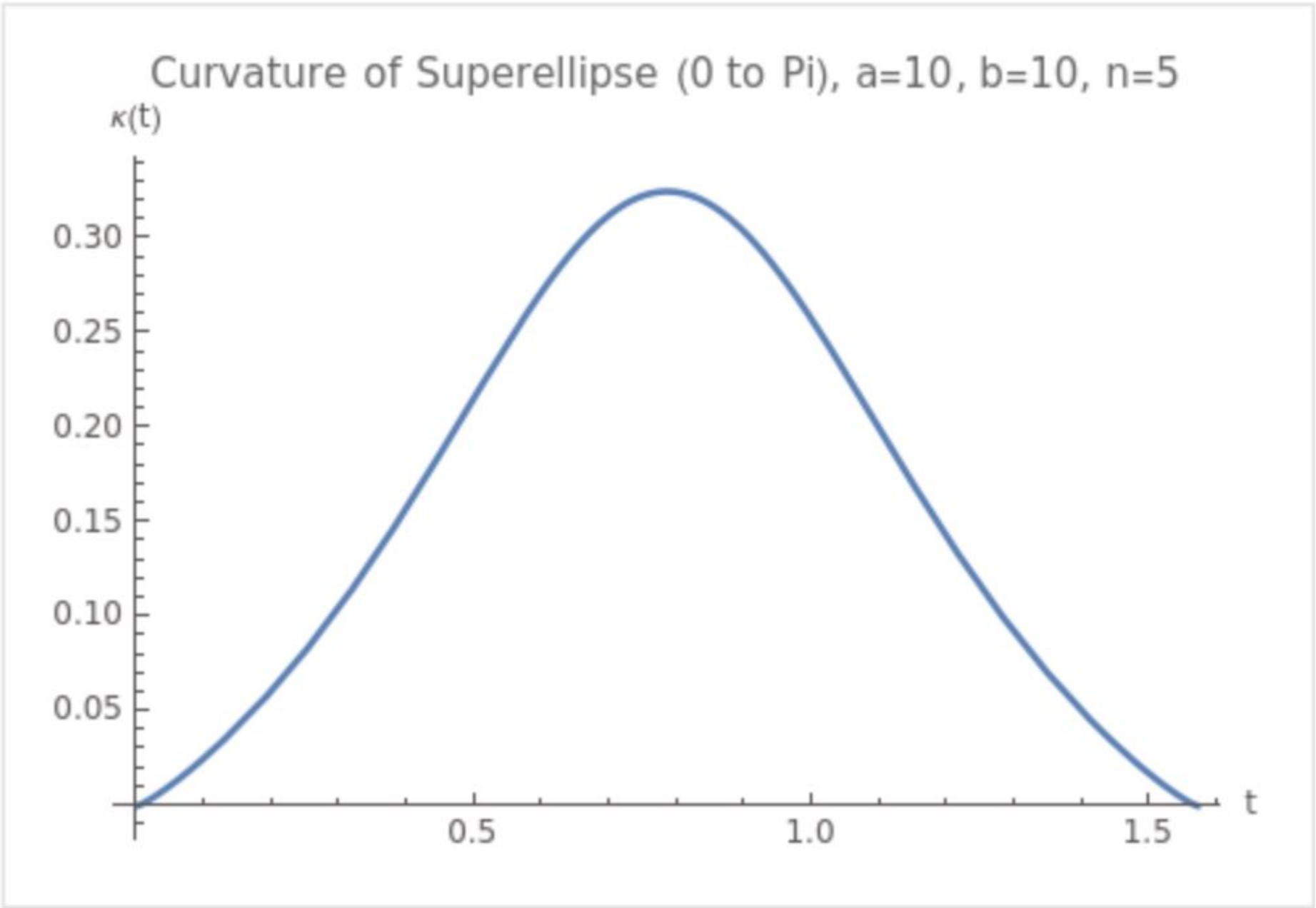
Rounded Square



Squircle

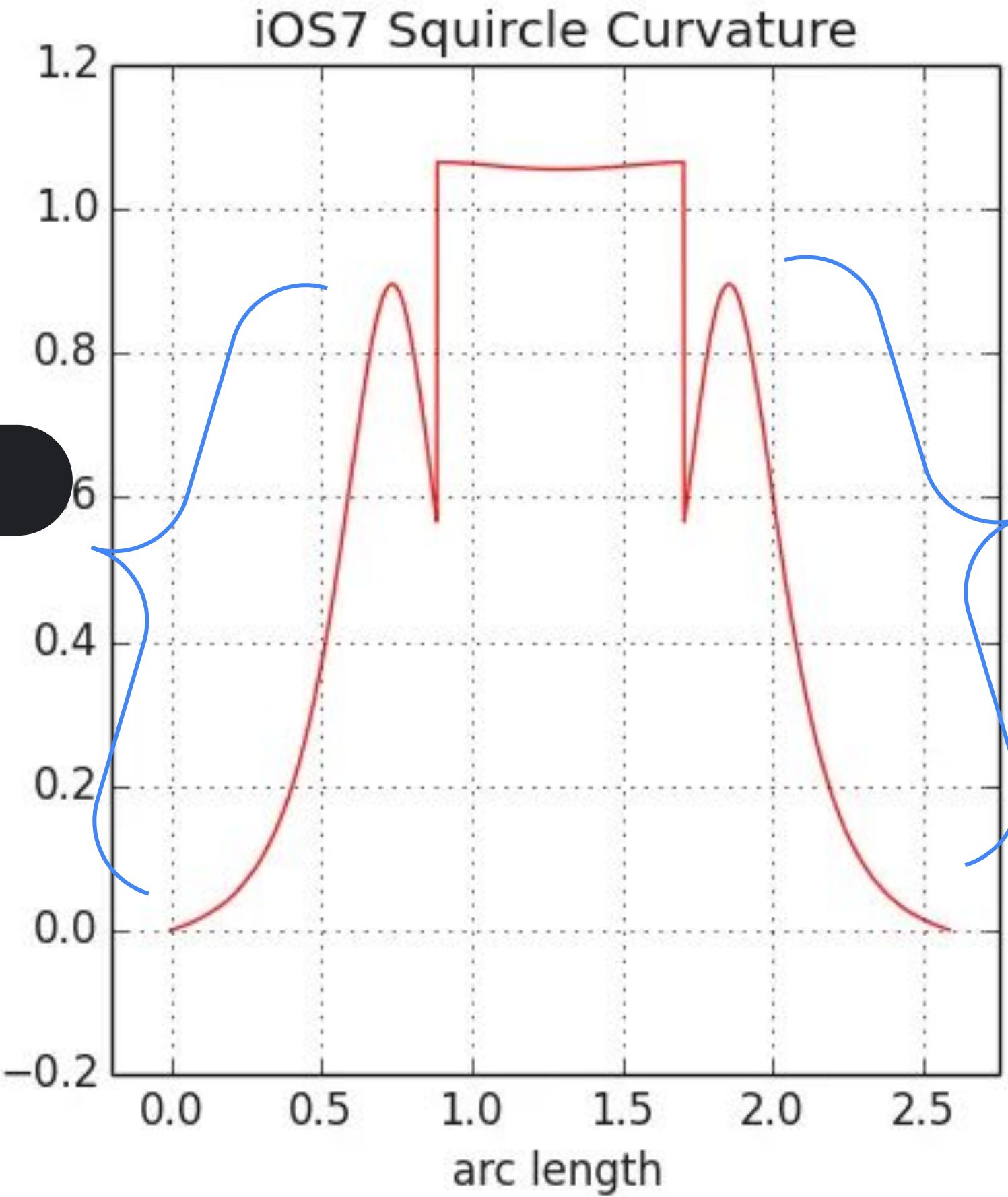
Squircle vs Superellipse

Superellipse's curvature



Figma's curvature analysis

Circle



Superellipse

Superellipse

RSuperellipse

Available in the stable channel today

Swift Package Manager

Supported on iOS and macOS

#4 - Melhorias para Android

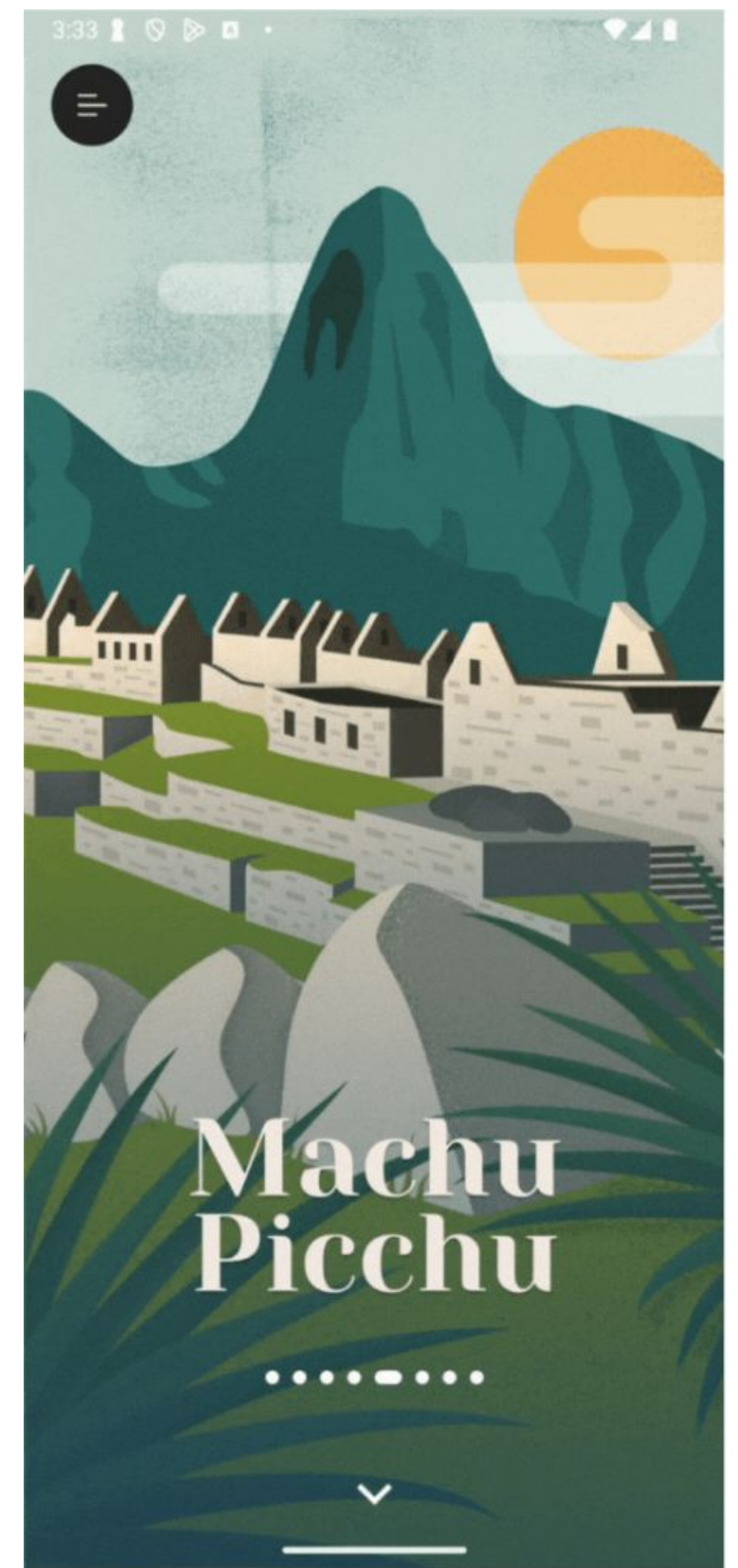
Impeller default on Android

Edge-to-edge

Without Edge-to-Edge



With Edge-to-Edge



Scribe

Scribe demo

|

#5 – Melhorias para WEB

WebAssembly

WebAssembly

Supported in 92% of top
web packages on pub.dev

Accessibility

Semantics roles

```
Semantics(  
  // Use `SemanticsRole.menuBar` if this is a custom menu bar.  
  role: SemanticsRole.menu,  
  child: CustomMenu(  
    children: [  
      MenuItemButton(onPressed: () {}, child: const Text('Menu item 0')),  
      MenuItemButton(onPressed: () {}, child: const Text('Menu item 1')),  
      SubmenuButton(  
        onPressed: () {},  
        child: const Text('Submenu 0'),  
      ),  
    ],  
  ),  
)
```


Semantics roles

```
Semantics(  
  // Use `SemanticsRole.menuBar` if this is a custom menu bar.  
  role: SemanticsRole.menu,  
  child: CustomMenu(  
    children: [  
      MenuItemButton(onPressed: () {}, child: const Text('Menu item 0')),  
      MenuItemButton(onPressed: () {}, child: const Text('Menu item 1')),  
      SubmenuButton(  
        onPressed: () {},  
        child: const Text('Submenu 0'),  
      ),  
    ],  
  ),  
)
```

Hot reload



Flutter Web

Flutter web with Hot Reload

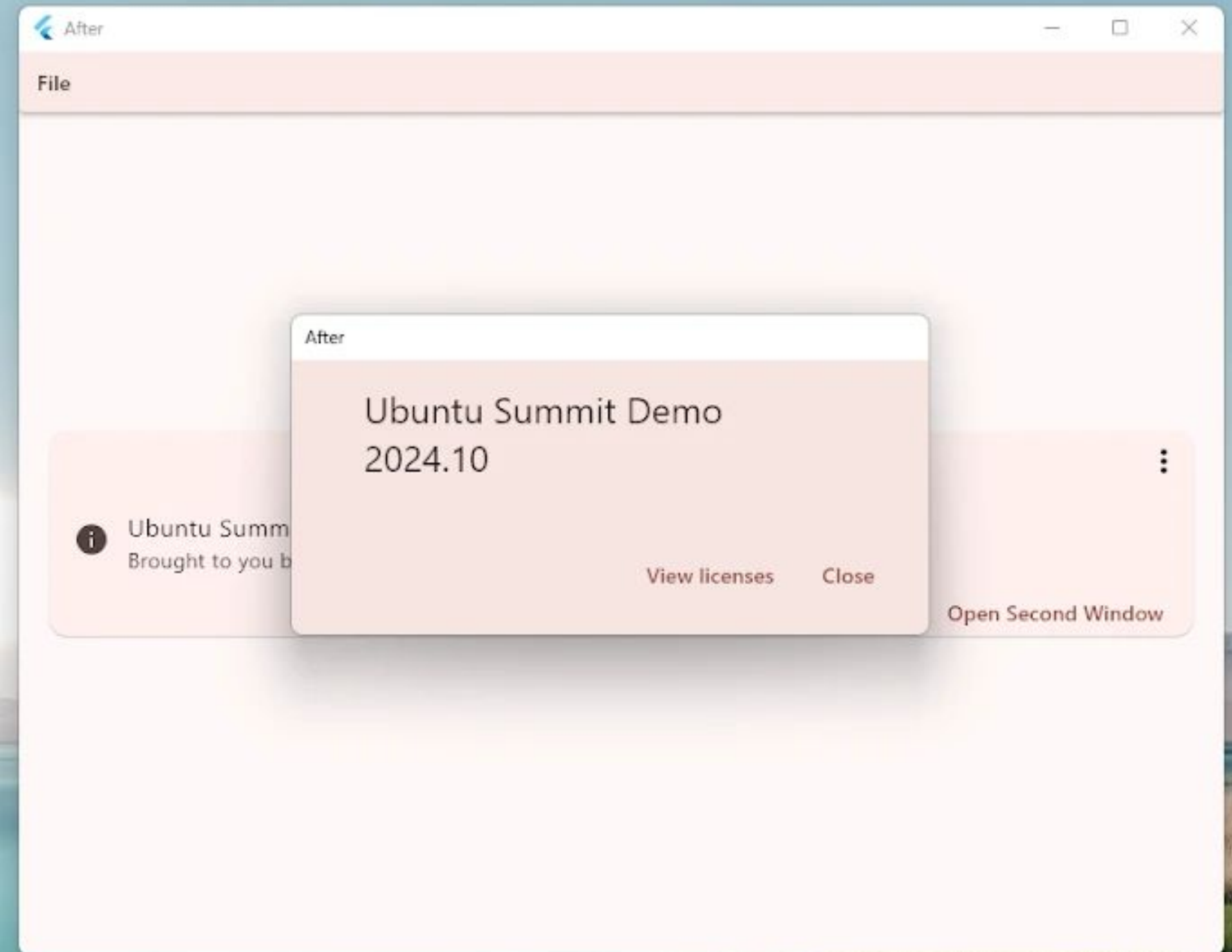
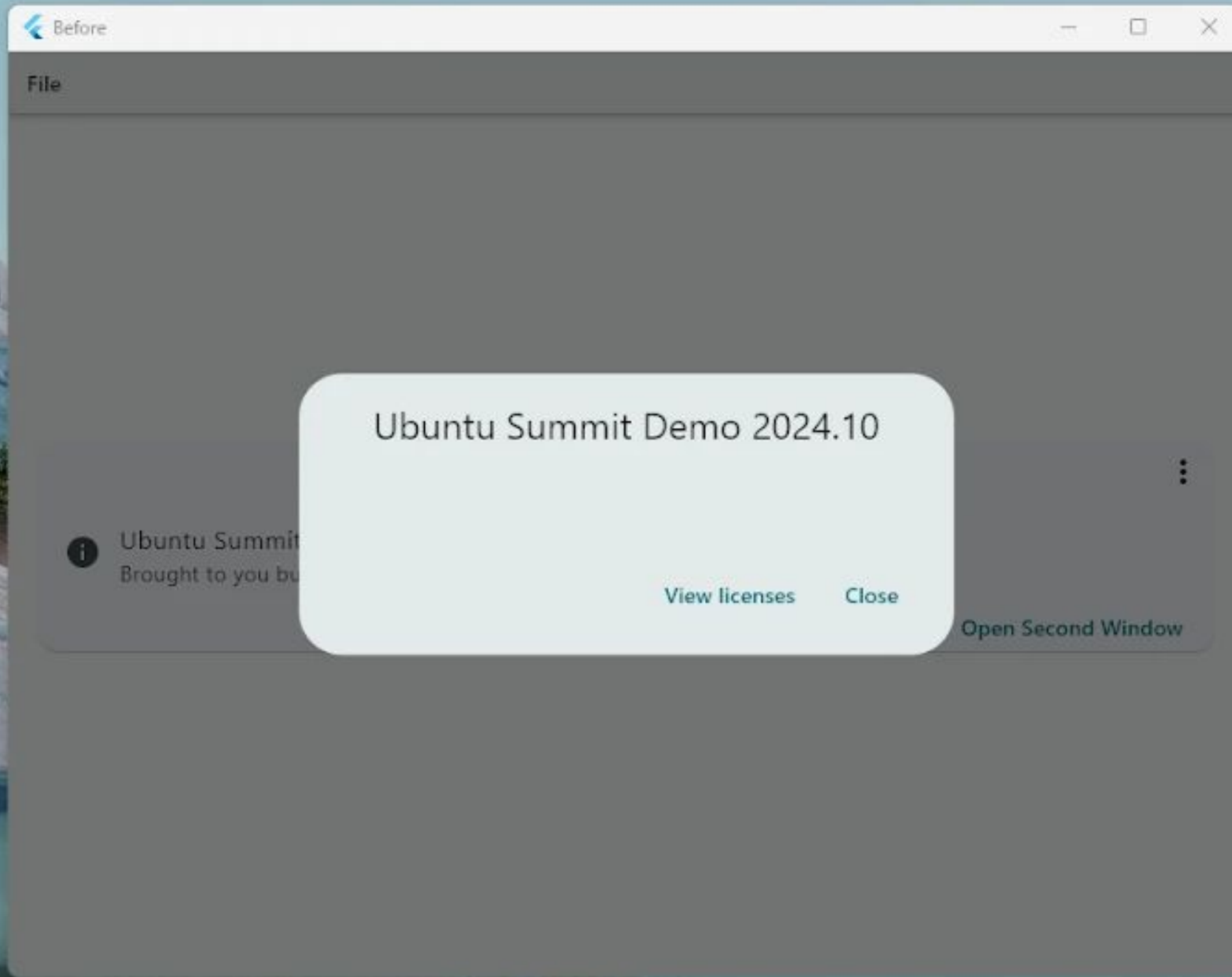
```
$ flutter run -d chrome
```


#6 – Ecosistema



Canonical

```
function filterStudies({ studies, filterByOrg = false, filterByLeadOrganizat
```





LG

```
function filterStudies({ studies, filterByOrg = false, filterByLeadOrganizat
```

Flutter :: webOS SDK

```
$ flutter-webos run -d tv --debug
Launching lib/main.dart on webOS in debug mode...
Uninstall com.flutter.app.helloworld from tv.
Success command
InstallApp : com.flutter.app.helloworld to tv.
Success command
runDebug Success
Launch Succeeded helloworld on tv
Syncing files to device webOS...
```

```
Flutter run key commands.
r Hot reload. 🔥🔥🔥
R Hot restart.
h List all available interactive commands.
d Detatch (terminate "flutter run" but leave application running).
c Clear the screen
q Quit (terminate the application on the device)
```

💪 Running with sound null safety 💪

- **Optimized performance:**
Native integration with webOS
- **Rich experiences:** Gamepad support, dynamic audio, Magic Remote, DRM Player, and more
- **Beta with leading partners:**
Lukas Klingsbo, Codemate, and more



SIGNS OF MAGIC

Start Game

Spell book

About



1080 motion™



GE APPLIANCES

#6 – Flutter e IA



I Am Developer 
@iamdeveloper

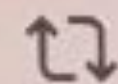


vibe coding, where 2 engineers can now
create the tech debt of at least 50 engineers

12:50 AM · 2025-03-20 · **571K** Views



169



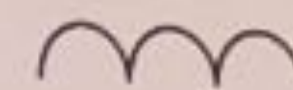
1.4K



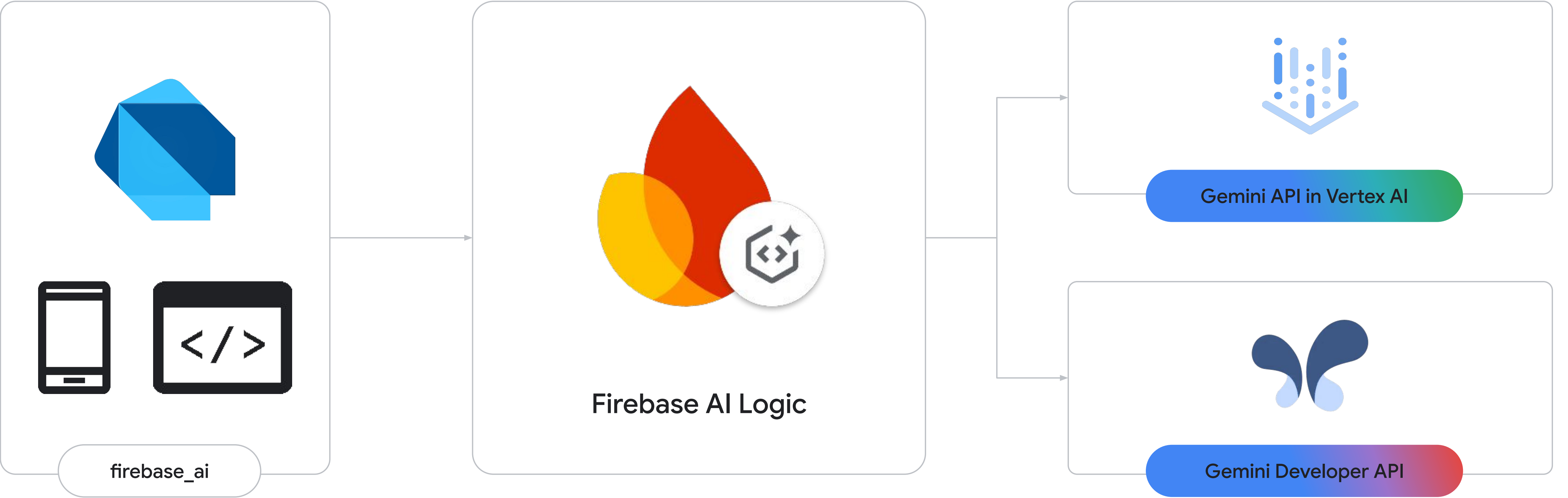
14K



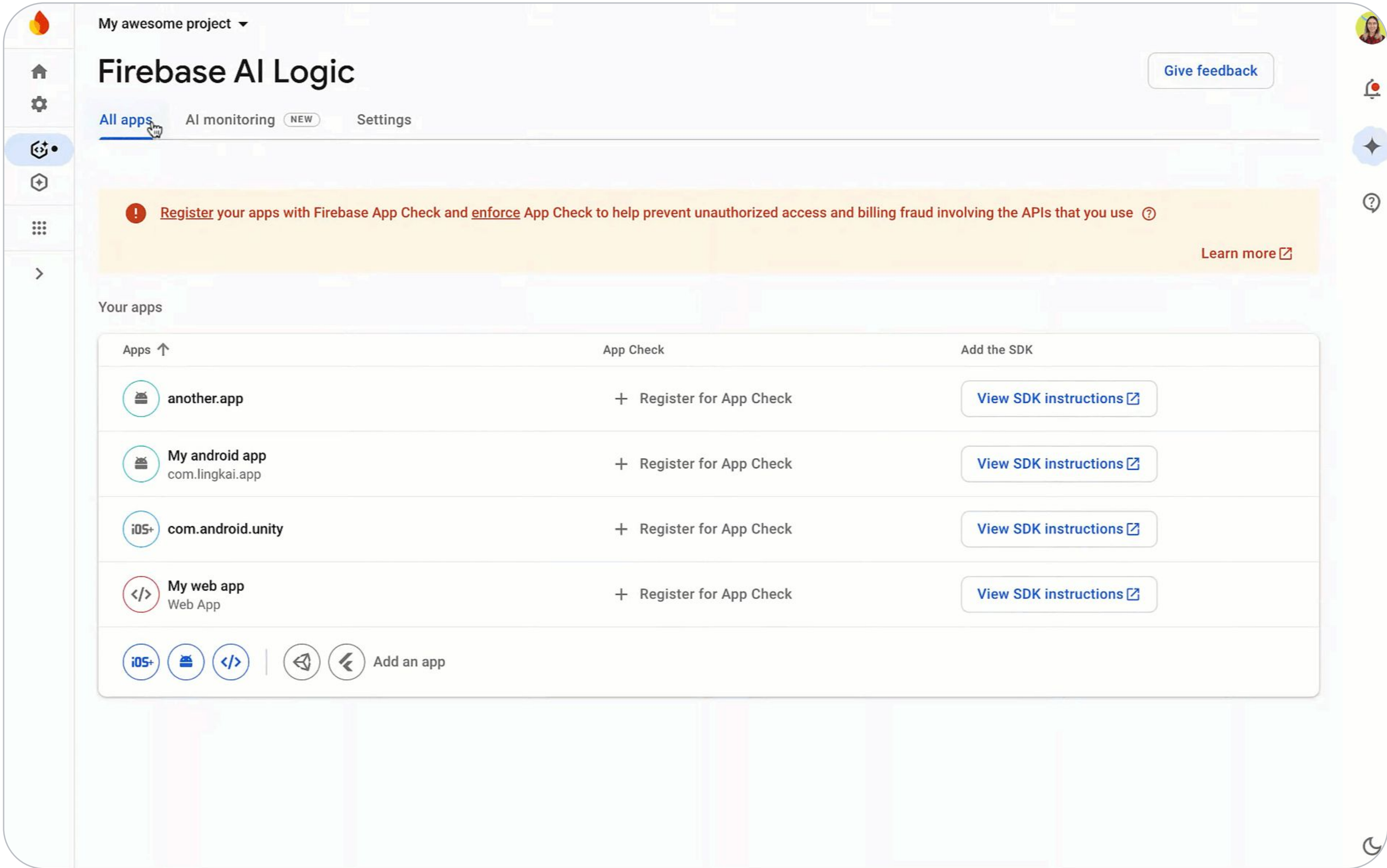
735



Firebase AI Logic

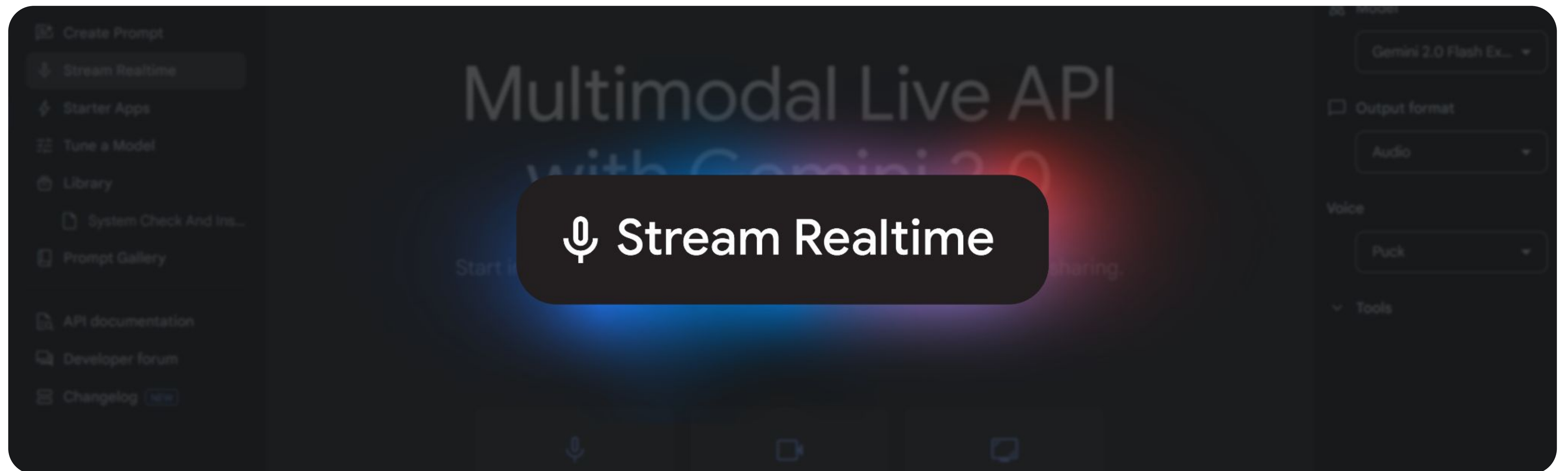


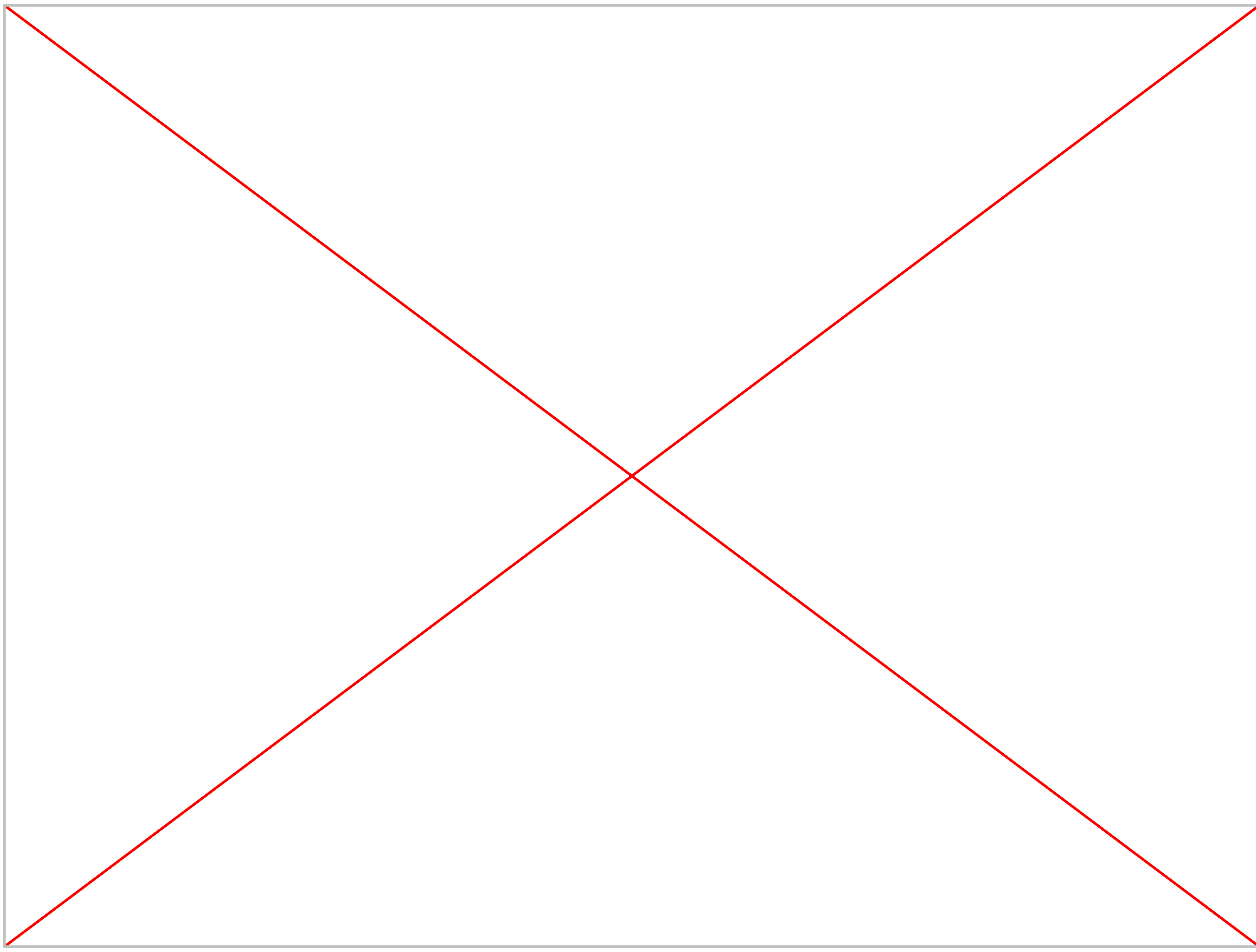
AI monitoring dashboard



Gemini Live API support

Baixa latência, interações bi-direcionais de voz e vídeo com Gemini





Vibe coding

Hello, Michael

Welcome back

Prototype an app with AI

An app that generates poems from photos

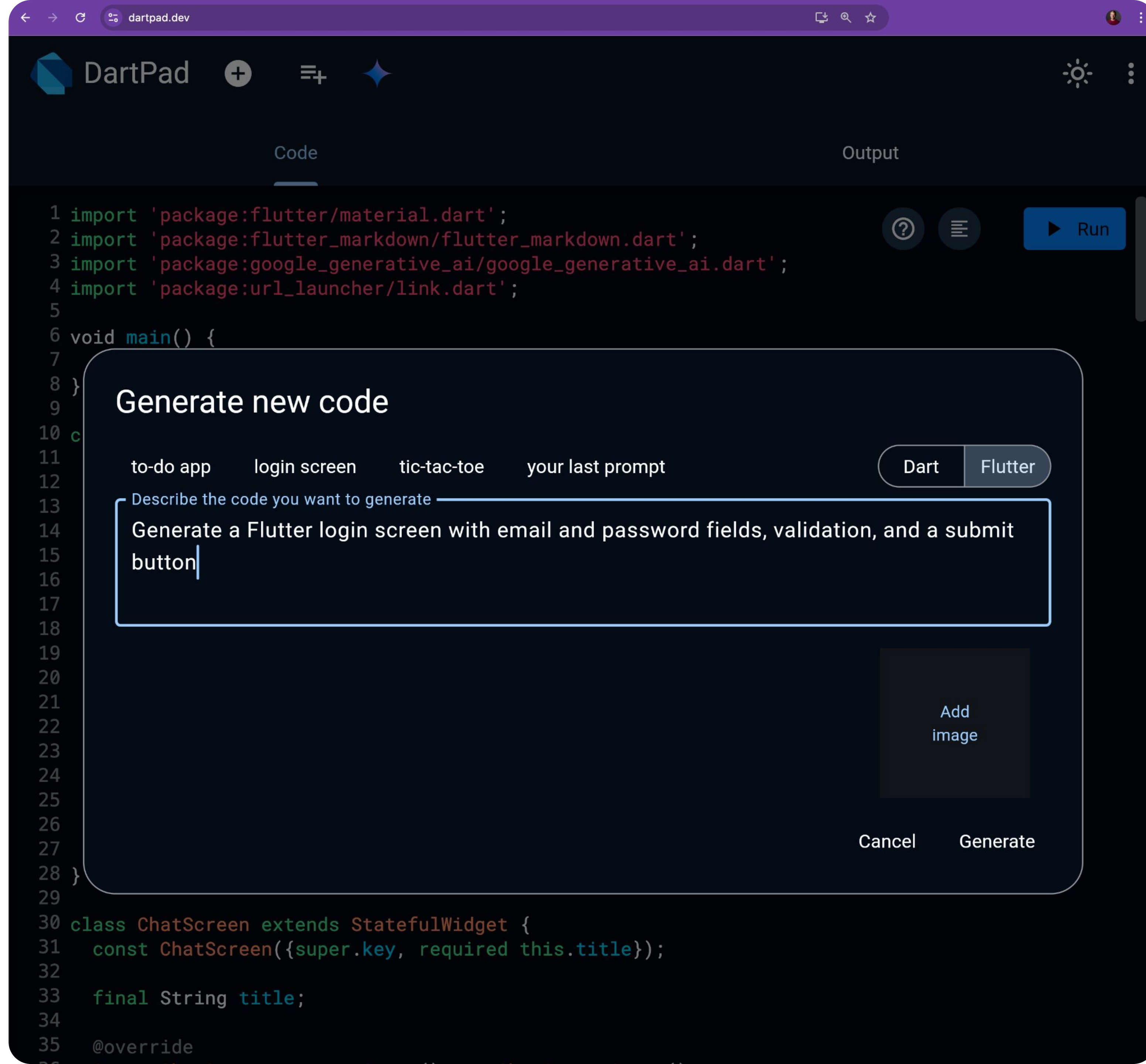
TAB



More sample prompts ▾

Gemini in DartPad

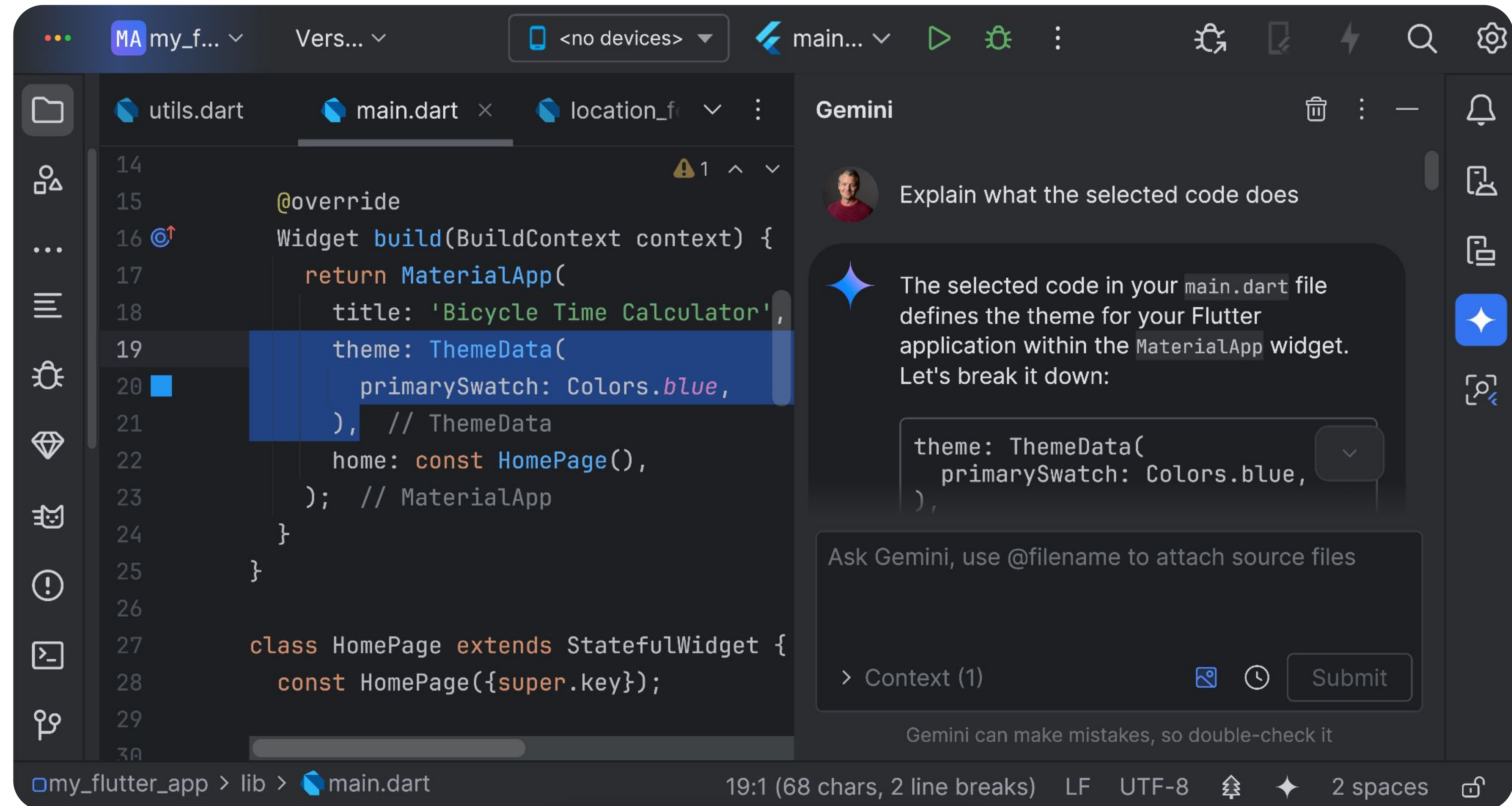
Powered by Google AI and the Gemini API.



Gemini in Android Studio

Announcing first-class support
for Dart and Flutter!

Available in the latest Android
Studio version, "Meerkat Feature
Drop" and later.



Gemini in Visual Studio Code

Available via the Gemini Code Assist extension.



Gemini Code Assist

[For Business](#)

[FAQ](#)

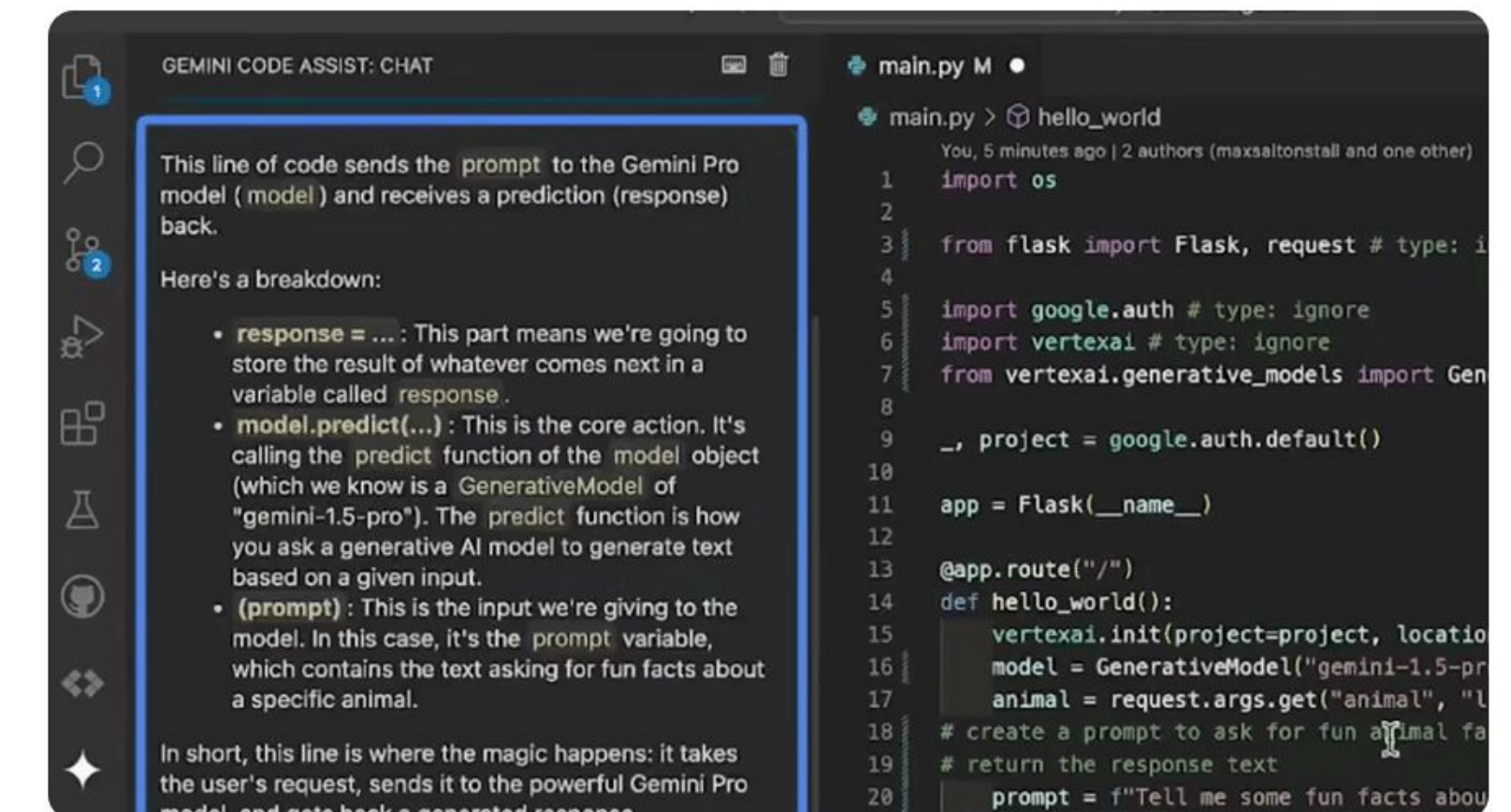
[Documentation](#)

Gemini Code Assist: AI coding assistance for any language

Uncap your potential and get all of your development done using Gemini Code Assist for individuals at no cost, or Gemini Code Assist for your business.

Use in Visual Studio Code

Use in JetBrains IDEs



Gemini in Android Studio for businesses

Optionally upgrade for enterprise-grade privacy and security



Your code stays secure

You retain complete control of your data and IP; we ensure your code, inputs, and responses aren't used for model training purposes.



Designed for Android

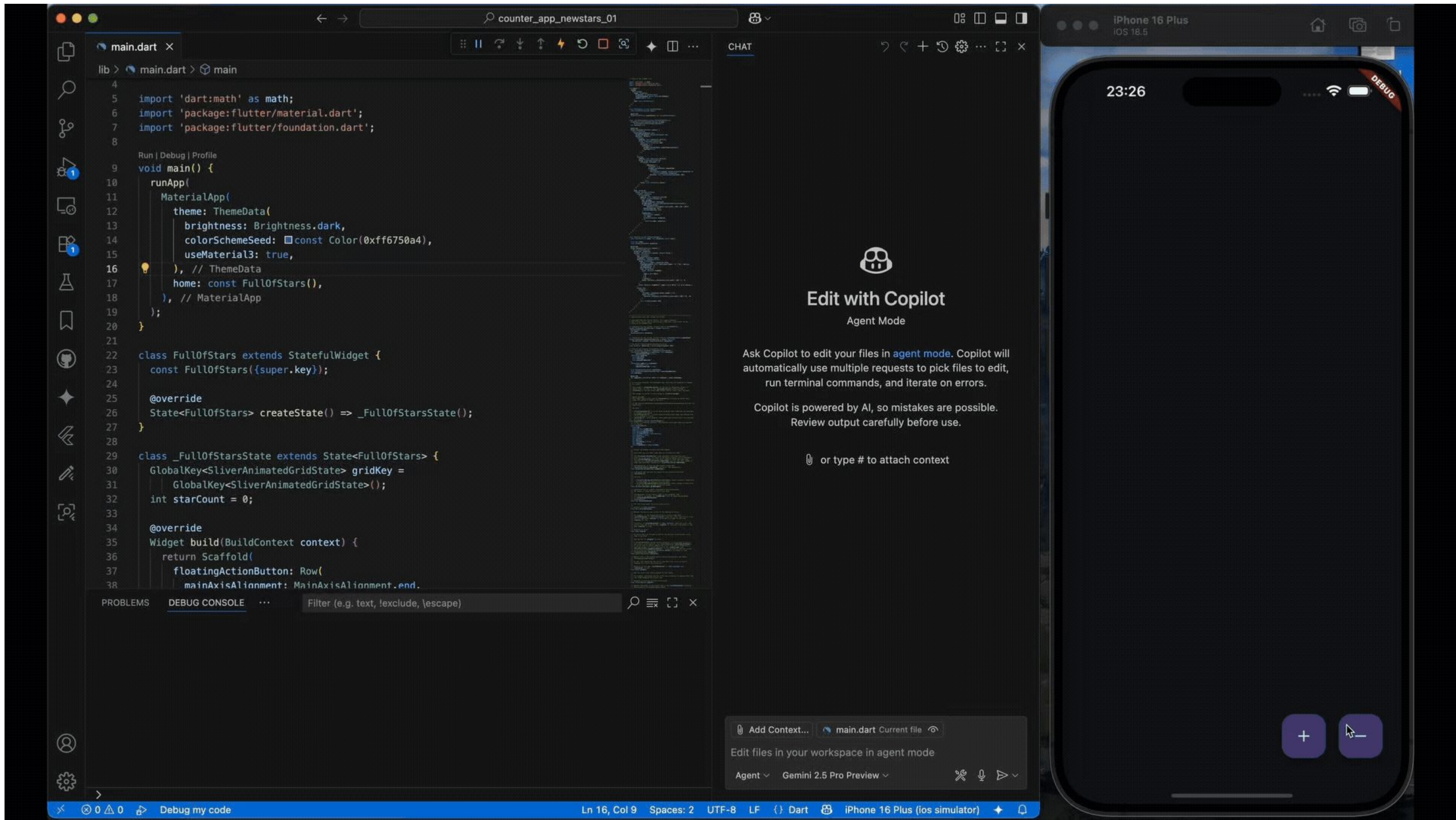
Build high-quality apps faster with smart tips and suggestions for Android. Get help with Compose UIs, Gradle build errors, and crash analytics—Gemini is designed specifically for the Android ecosystem.



Enterprise-grade management

Incorporate your company's best practices in your development workflow, manage access with admin controls, and use metrics to measure Gemini's impact on productivity.

MCP SERVERS



Vibe once, run everywhere



iOS



Android



Web



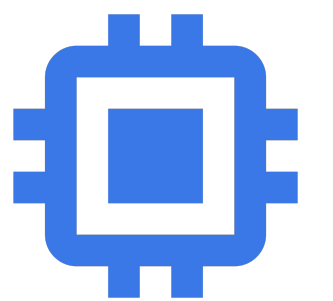
Windows



macOS



Linux



Embedded

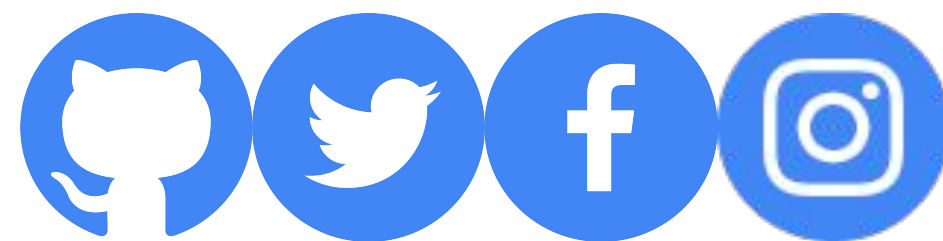
Thank You!



Toshi Ossada
GDE Flutter & Dart
Microsoft MVP
@toshiossada



youtube.flutterbrasil.com.br



[/toshiossada](#)